HORIZON EUROPE

Research and Innovation Action (RIA)

Resilient Mediterranean Agri-Food Systems Under Climate Uncertainty

TERRA-ADAPT

Field	Value
Call Identifier	HORIZON-CL6-2025-CLIMATE-01-01
Type of Action	RIA
Duration	48 months
TRL Range	TRL 3 to TRL 6
Total Budget	EUR 9,800,000
EU Contribution	EUR 9,800,000

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Abstract

TERRA-ADAPT addresses the accelerating climate threats facing Mediterranean agri-food systems — prolonged droughts, soil salinisation, erratic rainfall, and heat extremes — which jeopardise food security for over 150 million people. Current adaptation strategies are fragmented across plant genomics, precision agriculture, and socio-economic research. TERRA-ADAPT delivers the first integrated, closed-loop climate adaptation framework connecting multi-omics crop science, a real-time precision agriculture Decision Support System (DSS), and co-designed adaptation pathways, validated through 12 demonstration pilots in Spain, Italy, Greece, and Tunisia. Key outputs include: the largest open multi-omics dataset for Mediterranean crop adaptation (>50 TB, FAIR-compliant); a DSS fusing Copernicus Sentinel-2 imagery, 500+ IoT soil sensors, and seasonal climate forecasts (TRL 3 to 6); socio-economic resilience assessments of 1,200 farming families; and an open-access Mediterranean Agri-Climate Knowledge Platform with API access. The consortium of eight organisations — three universities, two research institutes, two SMEs, and a civil society organisation spanning six EU countries and Tunisia — will directly engage 25,000 farming households and demonstrate 15–20% crop yield loss reductions. TERRA-ADAPT contributes directly to the European Green Deal, Farm to Fork Strategy, and the EU Mission on Adaptation to Climate Change.
Keywords: Mediterranean agriculture, climate adaptation, food security, precision agriculture, decision support system, multi-omics, crop genomics, digital twin, living labs, Copernicus, IoT, socio-economic resilience, FAIR data, smallholder farmers, drought tolerance, salt tolerance, agri-food systems, climate-smart agriculture, open science, knowledge platform

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SECTION 1 — EXCELLENCE

1.1 Objectives and Ambition

The Mediterranean basin is among the most climate-vulnerable regions on Earth. Accelerating droughts, erratic precipitation patterns, soil salinisation, and recurrent heat extremes already threaten the food security of over 150 million people and are projected to intensify substantially by 2050. Despite this urgency, existing adaptation strategies remain fragmented: plant genomics, precision digital agriculture, and socio-economic resilience research each advance in disciplinary silos, generating knowledge that rarely translates into coherent, farmer-ready solutions. TERRA-ADAPT closes this gap by delivering the first fully integrated, closed-loop climate adaptation framework for Mediterranean agri-food systems, connecting genomic science, real-time digital tools, and co-designed adaptation pathways into a single operational system validated at landscape scale.

TERRA-ADAPT pursues five specific, measurable, and interlinked objectives. Objective 1: Develop climate-predictive phenotyping models for drought- and salt-tolerant varieties of wheat, olive, and tomato through multi-omics integration (genomics, transcriptomics, metabolomics) and AI-driven trait discovery, screening more than 2,000 accessions from national genebanks in Spain, Italy, Greece, and Tunisia. Objective 2: Design and validate a real-time precision agriculture Decision Support System (DSS) fusing Copernicus Sentinel-2 satellite imagery, an IoT soil sensor network of 500+ nodes, and seasonal climate forecasts, demonstrated across 12 pilot sites. Objective 3: Quantify the socio-economic resilience of smallholder farming communities across four Mediterranean sub-regions (Southern Spain, Southern Italy, Greece, Tunisia) through participatory living labs and household surveys covering at least 1,200 farming families. Objective 4: Co-create regionally tailored adaptation pathways with farmers, cooperatives, and policy-makers, validated through 12 demonstration pilots spanning the four target regions. Objective 5: Deliver an open-access Mediterranean Agri-Climate Knowledge Platform integrating all project datasets, models, and decision tools under FAIR principles with open API access for sustained post-project use.

The ambition of TERRA-ADAPT is twofold. Scientifically, it will produce the largest openly available multi-omics dataset for Mediterranean crop climate adaptation (>50 TB, FAIR-compliant) and advance machine learning methods for genotype-environment interaction prediction from TRL 3 to TRL 6. Societally, it will directly engage 25,000 farming households and demonstrate a measurable 15–20% reduction in crop yield losses at pilot sites. These objectives are pertinent to the call's focus on climate-resilient food systems, European Green Deal priorities, and the Farm to Fork Strategy target of reducing agriculture's vulnerability to climate shocks while maintaining productivity and sustainability.

1.2 Relation to the Work Programme

TERRA-ADAPT directly addresses the scope and expected outcomes of HORIZON-CL6-2025-CLIMATE-01-01, which calls for integrated, multi-actor research on climate adaptation in agri-food systems, with emphasis on the Mediterranean region, digital innovation, and co-creation with farming communities. The project responds specifically to the call's requirement to go beyond incremental adaptation by delivering systemic solutions that bridge genomics, precision agriculture, and socio-economic modelling.

The project aligns with multiple Horizon Europe strategic orientations: restoring Europe's ecosystems and managing natural resources sustainably; making Europe the first climate-neutral and sustainable economy through digital enablement; and creating resilient, inclusive societies. TERRA-ADAPT further complements the EU Mission on Adaptation to Climate Change by generating tested, deployable tools for farming communities. It is synergistic with ongoing EU investments in Copernicus Climate Change Service (C3S), AgriDataSpace, and the European Green Deal's soil health

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ambitions, while avoiding duplication by integrating rather than replicating these infrastructures. The inclusion of Tunisia as an associated country partner reflects the EU's commitment to neighbourhood policy and Euro-Mediterranean research cooperation under the Global Challenges pillar.

1.3 Concept and Methodology

The conceptual foundation of TERRA-ADAPT is a closed-loop adaptive management cycle operating across three interconnected layers: (i) the biological layer, where multi-omics data drive the identification of climate-resilient crop traits; (ii) the digital layer, where a DSS translates genomic and environmental data into actionable farm-level recommendations; and (iii) the social layer, where living labs and demonstration pilots validate these recommendations in real farming contexts and feed observations back to refine both breeding priorities and DSS algorithms. This architecture is novel because no existing programme integrates all three layers at the landscape scale required for Mediterranean agri-food systems.

In Work Package 1, multi-omics screening of 2,000+ crop accessions (wheat, olive, tomato) from national genebanks will be conducted using standardised phenotyping protocols under controlled drought and salinity stress conditions. Genomic, transcriptomic, and metabolomic datasets will be integrated using machine learning ensemble models — including gradient boosting, random forests, and deep learning architectures — to build genotype-environment interaction (GEI) prediction models capable of identifying trait combinations most relevant under projected 2050 climate scenarios. In Work Package 2, a modular DSS will be constructed around three data streams: Copernicus Sentinel-2 time-series imagery processed via Google Earth Engine for vegetation stress indices; an IoT sensor grid of 500+ nodes measuring soil moisture, salinity, and temperature in near-real time; and bias-corrected seasonal climate forecasts from Copernicus C3S. An agent-based model calibrated with household survey data from 1,200 farming families (WP3) will simulate farmer decision-making under climate uncertainty, providing a socio-economic layer to the DSS outputs. In Work Package 4, co-design workshops using the Multi-Actor Approach (MAA) will translate model outputs into 12 regionally tailored adaptation pathways, demonstrated through pilots across Southern Spain, Southern Italy, Greece, and Tunisia. The Knowledge Platform (WP5) will operate as the integration backbone, aggregating all datasets and tools under FAIR principles with open API access, ensuring interoperability with GEOSS, Copernicus, and national agricultural information systems.

Key methodological innovations include: (a) the first real-time closed-loop integration of crop genomics with a landscape-scale precision agriculture DSS; (b) an AI-driven 'digital twin' of Mediterranean farmland that dynamically updates as sensor and satellite data are ingested; and (c) a participatory co-design process that embeds social science throughout the technical development cycle rather than as an add-on. Gender disaggregation will be applied throughout all surveys and living lab activities to ensure the differential vulnerability and adaptive capacity of women farmers are captured. Open science principles — including pre-registration of research protocols, open-access publication, and FAIR data deposition from month 6 — are embedded by design rather than treated as a compliance requirement.

1.4 Interdisciplinary Considerations

TERRA-ADAPT is inherently interdisciplinary, requiring sustained collaboration across plant genomics, remote sensing, data science, agricultural engineering, environmental economics, social anthropology, and science communication. The consortium has been designed to ensure that each disciplinary domain is represented by a leading European or Mediterranean institution, and that genuine disciplinary integration — rather than parallel working — is achieved through joint work packages, shared data infrastructure, and co-authorship protocols. Specifically, WP3 (socio-economic modelling) and WP4 (demonstration pilots) serve as integration hubs where biological and digital outputs are tested against social realities. Monthly cross-WP technical meetings and a dedicated

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integration lead at Wageningen University & Research will ensure coherent knowledge flows across disciplinary boundaries. The Multi-Actor Approach, institutionalised through COPA-COGECA's coordination of WP6, ensures that farmer and civil society knowledge is treated as an epistemically valid input to the research process, not merely a dissemination target.

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SECTION 2 — IMPACT

2.1 Pathways Towards Impact

TERRA-ADAPT's pathway to impact operates at three interrelated levels. At the scientific level, the project will generate the largest open multi-omics dataset for Mediterranean crop climate adaptation (>50 TB, FAIR-compliant, deposited in ZENODO and EMBL-EBI), advance machine learning methods for GEI prediction, and produce 40+ peer-reviewed publications in high-impact journals. These outputs will directly feed into EU and national plant breeding programmes, with at least two crop variety breeding lines entering national seed certification processes by M48. At the technological level, the DSS will be validated through 12 demonstration pilots and licensed to at least three agri-tech companies by M42, creating a commercial pathway for sustained deployment. The Knowledge Platform will be sustained post-project through a partnership with GEOSS and Copernicus C3S, ensuring continued API access for national agricultural advisory services across the Mediterranean.

At the societal and policy level, TERRA-ADAPT will directly engage 25,000 farming households and demonstrate a 15–20% reduction in crop yield losses at pilot sites. Six or more policy briefs will be submitted to DG AGRI and DG CLIMA, and project representatives will participate in COP negotiations and EU Green Week policy roundtables. COPA-COGECA's institutional access to EU agricultural policy networks ensures that project findings reach decision-makers at the appropriate level and moment. The project's Mediterranean scope — spanning four sub-regions including an associated country — maximises transferability of results across the EU's most climate-exposed agricultural zones.

2.2 Dissemination, Exploitation and Communication

TERRA-ADAPT's dissemination strategy targets both scientific and practitioner audiences. At least 40 peer-reviewed publications will be produced over the project lifetime, with a minimum of 50% published as gold open access in journals aligned with Plan S requirements. An annual Mediterranean Agri-Climate Conference (M12, M24, M36, M48) will convene scientists, farmers, industry, and policy-makers to review results and foster cross-project learning. Practitioner guidebooks summarising adaptation pathways will be produced in five languages (English, Spanish, Italian, Greek, Arabic) and distributed through COPA-COGECA's network of national farmer organisations. A MOOC on climate-smart agriculture targeting extension officers and farming community leaders will be launched by M40, aiming for 3,000+ enrolments. A series of four short documentary films capturing farmer stories from the living labs will be produced and distributed via YouTube and Vimeo, targeting 50,000+ cumulative views by M48.

The exploitation strategy centres on two pathways. First, the DSS will be developed under a modular open-core architecture, allowing an open-source community edition to drive uptake while a supported commercial edition generates revenue through licensing agreements with agri-tech companies — with at least three agreements targeted by M42. Second, crop breeding lines developed under WP1 will be formally transferred to national seed certification programmes in Spain, Italy, Greece, and Tunisia through Material Transfer Agreements negotiated by M44. The Knowledge Platform's sustainability beyond the project lifetime will be secured through a formal partnership agreement with GEOSS and integration into the Copernicus C3S portfolio, ensuring ongoing data ingestion and user access without reliance on project funding. The exploitation roadmap (D6.4) will detail IP arrangements, revenue-sharing mechanisms, and SME commercialisation plans, developed jointly by AgriSense Technologies and CropOmics GmbH.

The communication plan targets broad public awareness and EU policy engagement. A professional project website will be live by M2, supported by active social media channels across LinkedIn, X (formerly Twitter), and Instagram, with content adapted for farming, scientific, and policy audiences. Policy roundtables will be organised at COP events and EU Green Week from M24 onwards, with

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targeted briefings delivered directly to DG AGRI and DG CLIMA officials. All communication materials will adhere to EC visual identity guidelines and carry the 'Funded by the European Union' statement.

2.3 Summary of Impact Indicators

Indicator	Target	Measurement
Peer-reviewed publications	≥40, of which ≥50% gold open access	Publication tracker and DOI registry updated quarterly
Farming households directly engaged	≥25,000 across four regions	Living lab and pilot site registration records
Crop yield loss reduction in pilot regions	15–20% relative to pre-project baseline	Agronomic trial data compared against site-specific baseline yield records
DSS licensing agreements signed	≥3 commercial agreements	Signed licensing contracts on file, reported in D6.4
Breeding lines entering national seed certification	≥2 national seed certification programmes	Material Transfer Agreement records and certification body confirmations
Knowledge Platform registered users	≥5,000 registered users by M48	Platform analytics dashboard (monthly reporting)
Policy briefs submitted to EU and national bodies	≥6 briefs submitted to DG AGRI, DG CLIMA, and national ministries	Submission confirmation receipts and acknowledgement letters
MOOC enrolments	≥3,000 enrolments by M48	MOOC platform registration data
Documentary film views	≥50,000 cumulative views across all films	YouTube and Vimeo analytics (public dashboards)
Open datasets published under FAIR principles	≥20 datasets totalling >50 TB	Repository DOIs registered in the project's FAIR data catalogue

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SECTION 3 — IMPLEMENTATION

3.1 Work Plan and Work Packages

WP	Title	Lead	PM	Duration
WP1	Climate-Adaptive Crop Genomics	Universidad de Córdoba	85	M1-M42
WP2	Precision Agriculture Decision Support System	CNR	78	M6-M44
WP3	Socio-Economic Resilience and Living Labs	Agricultural University of Athens	65	M4-M46
WP4	Demonstration Pilots and Adaptation Pathways	INRAT	70	M12-M48
WP5	Knowledge Platform and Data Integration	Wageningen University & Research	55	M1-M48
WP6	Dissemination, Exploitation and Communication	COPA-COGECA	30	M1-M48
WP7	Project Management and Coordination	Universidad de Córdoba	25	M1-M48

WP1: Climate-Adaptive Crop Genomics

Lead: Universidad de Córdoba | Person-months: 85 | Duration: M1 - M42

WP1 delivers multi-omics screening of 2,000+ crop accessions (wheat, olive, tomato) from national genebanks under simulated drought and salinity stress. Machine learning ensemble models are trained on integrated genomic, transcriptomic, and metabolomic datasets to build climate-predictive phenotyping models. CropOmics GmbH contributes high-throughput sequencing and bioinformatics pipelines. Outputs feed directly into WP2 (DSS trait modules) and WP4 (pilot variety selection).

T1.1: Genebank accession cataloguing and MTA negotiation — Compile and validate a catalogue of 2,000+ accessions from Spanish, Italian, Greek, and Tunisian national genebanks; negotiate Material Transfer Agreements by M3.

T1.2: Multi-omics phenotyping under stress conditions — Conduct controlled-environment phenotyping experiments combining genomics, transcriptomics, and metabolomics; deposit raw data in FAIR repository by M18.

T1.3: AI-driven GEI model development — Train and validate machine learning ensemble models for genotype-environment interaction prediction; produce climate-predictive phenotyping models by M24.

WP2: Precision Agriculture Decision Support System

Lead: CNR | Person-months: 78 | Duration: M6 - M44

WP2 designs, builds, and validates a modular DSS integrating Copernicus Sentinel-2 satellite data, an IoT soil sensor network (500+ nodes), and Copernicus C3S seasonal climate forecasts. AgriSense Technologies leads IoT deployment and sensor data pipelines. The DSS is validated in controlled environments by M24 and upgraded with full satellite integration by M36.

T2.1: IoT sensor network deployment — Deploy 500+ soil moisture, salinity, and temperature sensor nodes across pilot sites in four regions; establish real-time data transmission pipelines.

T2.2: DSS architecture design and Sentinel-2 integration — Specify modular DSS architecture; implement Sentinel-2 time-series processing pipelines for vegetation stress index computation.

T2.3: DSS prototype development and validation — Develop DSS prototype v1.0; conduct controlled-environment validation against agronomic benchmarks; release v2.0 with full satellite integration by M36.

WP3: Socio-Economic Resilience and Living Labs

Lead: Agricultural University of Athens | Person-months: 65 | Duration: M4 - M46

WP3 establishes participatory living labs in four regions, conducts household surveys with 1,200 farming families, and builds agent-based socio-economic resilience models calibrated to empirical survey data. Gender disaggregation is applied throughout. Living lab outputs feed back into WP1 breeding priorities and WP2 DSS parameterisation.

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T3.1: Living lab establishment and governance — Establish living labs in Southern Spain, Southern Italy, Greece, and Tunisia; develop governance frameworks and recruit farming family participants.
T3.2: Baseline socio-economic survey — Conduct household surveys covering 1,200 farming families; collect data on income, land use, climate exposure, and adaptive capacity with gender disaggregation.
T3.3: Agent-based resilience modelling — Develop and calibrate agent-based models of farmer decision-making under climate uncertainty; integrate outputs into the DSS socio-economic module.

WP4: Demonstration Pilots and Adaptation Pathways

Lead: INRAT | Person-months: 70 | Duration: M12 - M48

WP4 operates 12 demonstration pilots across four regions, implementing DSS-guided and genomics-informed crop management practices. Co-design workshops apply the Multi-Actor Approach to translate pilot results into regionally tailored adaptation pathway guidelines. AgriSense Technologies supports precision irrigation and sensor monitoring at pilot sites.

T4.1: Pilot site baseline monitoring — Establish agronomic and environmental baselines at all 12 pilot sites; install monitoring infrastructure and recruit farmer partners.
T4.2: DSS-guided pilot implementation — Deploy DSS recommendations at pilot sites; collect seasonal agronomic performance data against baseline; iterate recommendations using living lab feedback.
T4.3: Co-design of adaptation pathway guidelines — Conduct Multi-Actor Approach workshops with farmers, cooperatives, and policy-makers; synthesise lessons from pilots into regionally differentiated adaptation pathway documents.

WP5: Knowledge Platform and Data Integration

Lead: Wageningen University & Research | Person-months: 55 | Duration: M1 - M48

WP5 develops and operates the Mediterranean Agri-Climate Knowledge Platform, integrating all project datasets, models, and tools under FAIR principles with open API access. Wageningen leads data governance, interoperability standards, and platform sustainability planning. The platform serves as the integration backbone for all WPs and will be sustained post-project through GEOSS and Copernicus C3S partnerships.

T5.1: FAIR data management and repository infrastructure — Develop the project Data Management Plan; establish FAIR-compliant data repositories and metadata standards from M1; deposit first datasets by M6.
T5.2: Knowledge Platform development and API release — Design and build the platform front-end, back-end, and open API; release v1.0 with full API documentation by M30.
T5.3: Platform sustainability and partnership agreements — Negotiate integration agreements with GEOSS and Copernicus C3S; develop and publish the platform sustainability plan by M44.

WP6: Dissemination, Exploitation and Communication

Lead: COPA-COGECA | Person-months: 30 | Duration: M1 - M48

WP6 coordinates all dissemination, exploitation, and communication activities. COPA-COGECA leverages its European network of farmer organisations for practitioner engagement, policy roundtables, and guidebook distribution. SME partners AgriSense and CropOmics lead exploitation activities including DSS licensing and IP management.

T6.1: Dissemination and communication plan — Develop a comprehensive dissemination and communication plan covering target audiences, channels, key messages, and timelines; publish by M4.
T6.2: Practitioner engagement and guidebook production — Produce practitioner guidebooks in five languages; coordinate annual Mediterranean Agri-Climate Conference; produce and distribute four documentary films.
T6.3: Exploitation and commercialisation — Develop exploitation roadmap; lead DSS licensing negotiations with agri-tech companies; oversee Material Transfer Agreements for crop breeding lines; launch MOOC by M40.

WP7: Project Management and Coordination

Lead: Universidad de Córdoba | Person-months: 25 | Duration: M1 - M48

WP7 ensures efficient scientific, financial, and administrative management of the project. The Coordinator establishes governance structures, manages reporting to the European Commission, ensures ethical compliance including GDPR, and operates the internal communication and quality assurance system.

T7.1: Governance and management infrastructure — Establish the Project Management Board, Scientific Advisory Board, and Stakeholder Advisory Panel; develop the project management handbook by M3.
T7.2: Financial, ethical, and reporting management — Manage financial reporting, audit preparation, and ethical oversight including GDPR compliance; submit periodic and final reports to the EC on schedule.

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T7.3: Quality assurance and risk monitoring — Operate the quality assurance system; maintain the risk register; escalate and resolve emerging issues through the Project Management Board.

3.2 Consortium Composition

Organisation	Country	Type	Role	Description
Universidad de Córdoba	Spain	Public University	Coordinator	Project coordinator; leads WP1 (Climate-Adaptive Crop Genomics) and WP7 (Project Management). Brings expertise in plant genomics, phenotyping, and Mediterranean agronomy.
CNR (Consiglio Nazionale delle Ricerche)	Italy	Research Organisation	Partner	Leads WP2 (Precision Agriculture DSS). Contributes remote sensing, IoT systems, and data integration expertise.
Agricultural University of Athens	Greece	Public University	Partner	Leads WP3 (Socio-Economic Resilience and Living Labs). Contributes agricultural economics, participatory research, and Mediterranean farming systems expertise.
INRAT (Institut National de la Recherche Agronomique de Tunisie)	Tunisia	Research Institute (Associated Country)	Partner	Leads WP4 (Demonstration Pilots and Adaptation Pathways). Provides access to Tunisian genebanks, pilot sites, and farmer networks.
Wageningen University & Research	Netherlands	Research University	Partner	Leads WP5 (Knowledge Platform and Data Integration). Contributes data science, FAIR data governance, and platform engineering expertise.
AgriSense Technologies	Spain	SME	Partner	Contributes to WP2 and WP4. Provides IoT sensor hardware, field deployment logistics, and precision irrigation technology for pilot sites.
CropOmics GmbH	Germany	SME	Partner	Contributes to WP1. Provides high-throughput

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				sequencing services, bioinformatics pipelines, and AI-driven trait discovery tools.
COPA-COGECA	Belgium	Civil Society Organisation	Partner	Leads WP6 (Dissemination, Exploitation and Communication). Leverages its network of European and Mediterranean farmer organisations for practitioner engagement, guidebook distribution, and policy roundtables.

The TERRA-ADAPT consortium brings together a complementary and non-overlapping set of competencies that no single institution could replicate. The three universities (Córdoba, Athens, Wageningen) cover the full spectrum from genomics to socio-economics to data science. The two research organisations (CNR, INRAT) provide operational field infrastructure and regional embedding in Italy and Tunisia respectively. The two SMEs (AgriSense Technologies, CropOmics GmbH) inject commercial-grade technology and exploitation capacity, ensuring that research outputs have a viable path to market. COPA-COGECA provides unrivalled access to the end-user community of European and Mediterranean farmers, guaranteeing that co-design and dissemination activities reach their intended beneficiaries. The consortium spans six EU Member States plus Tunisia, providing the geographical coverage necessary for meaningful Mediterranean-scale analysis and maximising the transferability of project results across the EU's most climate-exposed agricultural regions.

3.3 Milestones

ID	Title	WP	Due	Verification
MS1	Multi-omics screening of 2,000+ crop accessions complete	WP1	M18	Full multi-omics dataset deposited in FAIR-compliant repository with registered DOI; completion confirmed by WP1 leader report
MS2	DSS prototype validated in controlled environment	WP2	M24	Validation report submitted to Project Management Board, including performance metrics against pre-defined benchmarks
MS3	Living labs operational in all four target regions	WP3	M12	Governance documents signed, baseline surveys initiated, and regional living lab coordinators confirmed in all four regions
MS4	All 12 demonstration pilots launched	WP4	M24	Pilot site monitoring data active and partner farmer confirmation letters

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				received for all 12 sites
MS5	Knowledge Platform v1.0 live with open API	WP5	M30	Platform URL publicly accessible; API documentation published and tested; confirmed by Wageningen technical report
MS6	Final integrated cross-WP assessment complete	WP3, WP4	M46	Integrated assessment report synthesising genomics, DSS performance, pilot agronomic results, and socio-economic modelling submitted to PMB
MS7	DSS licensing negotiations initiated with agri-tech companies	WP6	M36	Letters of intent from at least two agri-tech companies received and filed; reported in periodic progress report
MS8	Policy briefs submitted to DG AGRI and DG CLIMA	WP6	M42	Submission confirmation receipts from DG AGRI and DG CLIMA on file; copies of submitted briefs annexed to progress report

3.4 Deliverables

ID	Title	WP	Due	Type	Dissemination
D1.1	Genebank accession catalogue	WP1	M6	Report	Public
D1.2	Multi-omics dataset (>50 TB, FAIR-compliant)	WP1	M18	FAIR Dataset	Public
D1.3	Climate-predictive phenotyping models	WP1	M24	Software/Report	Public
D2.1	IoT sensor network deployment report	WP2	M12	Report	Public
D2.2	DSS architecture specifications	WP2	M12	Report	Public
D2.3	DSS prototype v1.0	WP2	M24	Software	Confidential
D2.4	DSS v2.0 with full Sentinel-2 satellite integration	WP2	M36	Software	Public
D3.1	Living lab establishment report	WP3	M8	Report	Public
D3.2	Socio-economic	WP3	M18	Report	Public

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	resilience baseline report				
D3.3	Agent-based socio-economic resilience model	WP3	M36	Software/Report	Public
D4.1	Pilot site baseline monitoring report	WP4	M18	Report	Public
D4.2	Interim pilot results report	WP4	M36	Report	Public
D4.3	Regional adaptation pathway guidelines	WP4	M42	Report	Public
D5.1	FAIR data management plan	WP5	M6	ORDP	Public
D5.2	Knowledge Platform v1.0 with open API	WP5	M36	Software	Public
D5.3	Platform sustainability plan	WP5	M44	Report	Public
D6.1	Dissemination and communication plan	WP6	M4	Report	Public
D6.2	Practitioner guidebooks in five languages	WP6	M40	Report	Public
D6.3	MOOC on climate-smart agriculture	WP6	M40	Other	Public
D6.4	Exploitation roadmap	WP6	M44	Report	Confidential
D7.1	Project management handbook	WP7	M3	Report	Confidential
D7.2	Data ethics and GDPR compliance report	WP7	M6	Ethics Report	Confidential
D7.3	Final project report	WP7	M48	Report	Public

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3.5 Resources and Budget Summary

Cost Category	Amount (EUR)
Personnel Costs	5880000
Subcontracting	490000
Travel & Subsistence	392000
Equipment	588000
Other Direct Costs	588000
Indirect Costs (25%)	1862000
TOTAL	9800000

3.6 Risk Management

ID	Risk	Likelihood	Impact	Mitigation

R1	Sensor network failure or persistent data gaps in remote pilot sites due to hardware malfunction, connectivity issues, or environmental damage	Medium	High	Redundant sensor deployment at critical nodes; satellite-derived gap-filling algorithms (Sentinel-2 and ERA5 reanalysis) to compensate for in-situ data loss; local technician training and preventive maintenance schedules established from M6
R2	Low farmer participation or dropout in living labs, particularly in Tunisia and Greece, due to competing demands on farmer time, distrust of research processes, or cultural barriers	Medium	High	COPA-COGECA farmer mobilisation through established national farmer organisation networks; financial compensation for farmer participation time; culturally adapted engagement protocols co-designed with local partners from M4; replacement participant lists maintained
R3	Delays in genebank accession access due to regulatory barriers, phytosanitary restrictions, or slow Material Transfer Agreement negotiations	Low	Medium	MTA negotiations initiated in M1–M3 in parallel with project launch; backup accession lists prepared covering 20% additional accessions; INRAT and Córdoba genebank offices act as co-facilitators for cross-border access
R4	DSS integration complexity exceeding planned timeline, particularly in achieving seamless data fusion across Sentinel-2, IoT, and climate forecast data streams	Medium	Medium	Modular DSS architecture with decoupled components allowing partial deployment if full integration is delayed; agile sprint development cycles with fortnightly reviews; dedicated integration lead at Wageningen University & Research overseeing cross-stream data harmonisation
R5	Political or institutional instability affecting the Tunisian partner	Low	High	Dual Principal Investigator structure at INRAT with clearly defined succession;

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	(INRAT), potentially disrupting WP4 pilot activities and data collection in the Tunisian sub-region			10% budget contingency at coordinator level; Universidad de Córdoba designated as backup lead for WP4 Tunisian activities; critical data backed up to EU-hosted repositories monthly

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